$$\therefore Ab_2 = \begin{bmatrix} b_{12}+b_{22}+b_{32} \\ b_{12}+2b_{22}+3b_{32} \\ b_{12}+4b_{22}+5b_{32} \end{bmatrix}$$ (Similar calculation as Ab_1)

$$Ab_3 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 5 \end{bmatrix} \begin{bmatrix} b_{13} \\ b_{23} \\ b_{33} \end{bmatrix} = \begin{bmatrix} b_{13}+b_{23}+b_{33} \\ b_{13}+2b_{23}+3b_{33} \\ b_{13}+4b_{23}+5b_{33} \end{bmatrix}$$ (Similar calculation as Ab_1)

$$\therefore AB = \begin{bmatrix} b_{11}+b_{21}+b_{31} & b_{12}+b_{22}+b_{32} & b_{13}+b_{23}+b_{33} \\ b_{11}+2b_{21}+3b_{31} & b_{12}+2b_{22}+3b_{32} & b_{13}+2b_{23}+3b_{33} \\ b_{11}+4b_{21}+5b_{31} & b_{12}+4b_{22}+5b_{32} & b_{13}+4b_{23}+5b_{33} \end{bmatrix}$$

$$BA = \begin{bmatrix} Ba_1 & Ba_2 & Ba_3 \end{bmatrix}$$

$$Ba_1 = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 1\begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix} + 1\begin{bmatrix} b_{12} \\ b_{22} \\ b_{32} \end{bmatrix} + 1\begin{bmatrix} b_{13} \\ b_{23} \\ b_{33} \end{bmatrix}$$ (By defn. of Ax)

$$= \begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix} + \begin{bmatrix} b_{12} \\ b_{22} \\ b_{32} \end{bmatrix} + \begin{bmatrix} b_{13} \\ b_{23} \\ b_{33} \end{bmatrix}$$ (By defn. of scalar multiple)

$$= \begin{bmatrix} b_{11}+b_{12}+b_{13} \\ b_{21}+b_{22}+b_{23} \\ b_{31}+b_{32}+b_{33} \end{bmatrix}$$ (By defn. of vector sum)

$$Ba_2 = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} = 1\begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix} + 2\begin{bmatrix} b_{12} \\ b_{22} \\ b_{32} \end{bmatrix} + 4\begin{bmatrix} b_{13} \\ b_{23} \\ b_{33} \end{bmatrix}$$ (By defn. of Ax)

$$= \begin{bmatrix} b_{11}+2b_{12}+4b_{13} \\ b_{21}+2b_{22}+4b_{23} \\ b_{31}+2b_{32}+4b_{33} \end{bmatrix}$$ (Similar calculation as Ba_1)

$$Ba_3 = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} = 1\begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix} + 3\begin{bmatrix} b_{12} \\ b_{22} \\ b_{32} \end{bmatrix} + 5\begin{bmatrix} b_{13} \\ b_{23} \\ b_{33} \end{bmatrix}$$ (By defn. of Ax)

$$= \begin{bmatrix} b_{11}+3b_{12}+5b_{13} \\ b_{21}+3b_{22}+5b_{23} \\ b_{31}+3b_{32}+5b_{33} \end{bmatrix}$$ (Similar calculation as Ba_1)

$$BA = \begin{bmatrix} b_{11}+b_{12}+b_{13} & b_{11}+2b_{12}+4b_{13} & b_{11}+3b_{12}+5b_{13} \\ b_{21}+b_{22}+b_{23} & b_{21}+2b_{22}+4b_{23} & b_{21}+3b_{22}+5b_{23} \\ b_{31}+b_{32}+b_{33} & b_{31}+2b_{32}+4b_{33} & b_{31}+3b_{32}+5b_{33} \end{bmatrix}$$